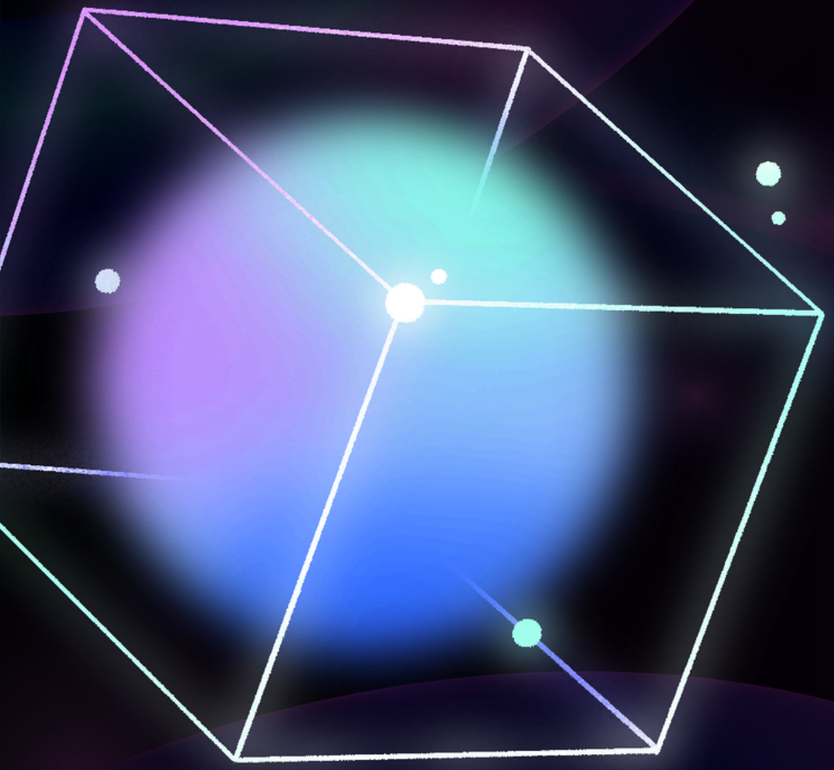
DEEP
NEURAL
NETWORKS

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DUAL-INPUT EMOTION CLASSIFICATION IN K-POP LYRICS

CNN VS. LINEARSVC BASELINES

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ABSTRACT & MOTIVATION

PROJECT SCOPE & BILINGUAL CHALLENGE

Analyzing bilingual Korean-English K-Pop lyric datasets (bts.csv) while filtering out Japanese tracks and missing text.

ACOUSTIC LABEL GROUNDING (THAYER'S MODEL)

Addressed random noise and subjective bias by grounding target labels into 4 affective states (Happy, Sad, Raring, Calm) using Spotify valence and energy thresholds.

DEFENSIBLE SPLIT & PREPROCESSING

Implemented an 80/20 stratified train-test split (random_state=42) with standard scaling on engineered metadata to prevent data leakage.

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LITERATURE REVIEW

AUTHOR	REVIEW
Alvitodinova et al. (2025)	Evaluated logistic regression, BiLSTM-XGBoost stacking, and DistilBERT for music lyric emotion classification across Sadness, Tenderness, and Tension categories
Miraz et al. (2025)	Explored emotion detection in music lyrics by comparing classical ML models (Random Forest, SVM, LightGBM) against deep learning architectures (ANN, CNN, LSTM, BiLSTM), achieving robust emotion classification (~78.3 accuracy).
Tang et al. (2023)	Investigated multimodal digital music emotion classification using audio-text feature fusion and CNN-SVM classifiers under Thayer’s emotion framework (>72% accuracy).

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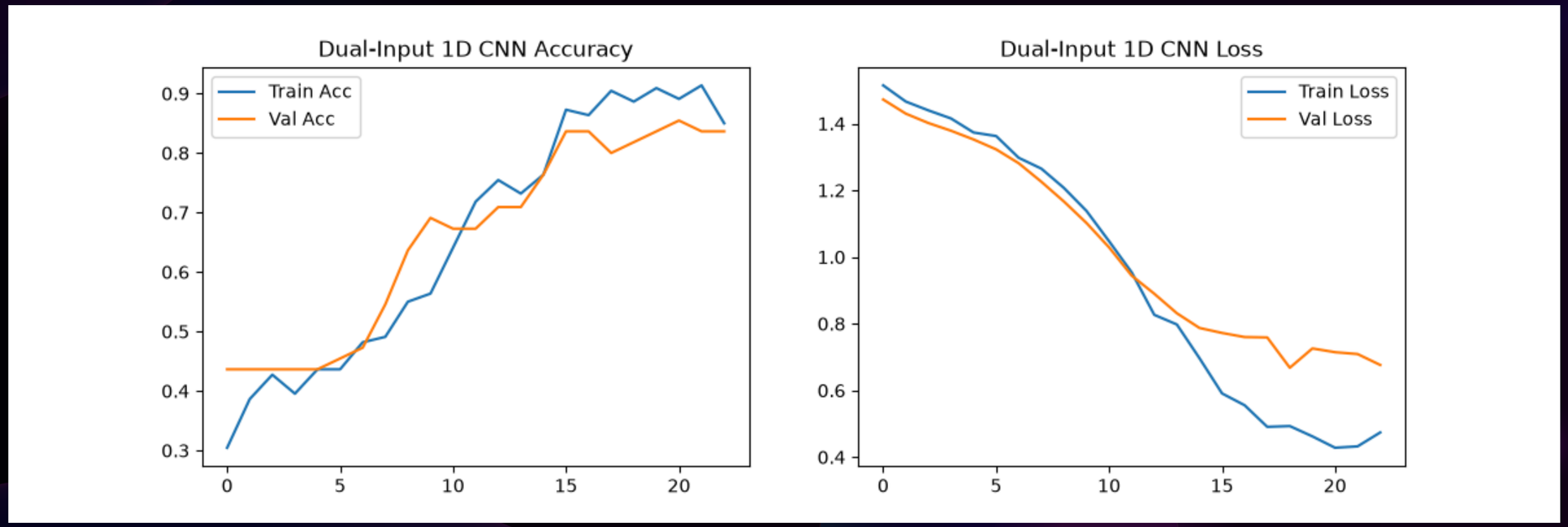
MODEL PERFORMANCE & TIMING

MODEL ARCHITECTURE	ACCURACY	PRECISION	RECALL	F1-SCORE	TRAIN TIME (S)	INTERFERENCE (S)
TF-IDF + LINEARSVC (BASELINE)	0.8545	0.861	0.85	0.854	0.118	0.001
DUAL-INPUT 1D CNN (REGULARIZED)	0.8182	0.795	0.809	0.798	4.319	0.166

MODEL
COMPARISON

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LEARNING CURVES & CONFUSION MATRICES

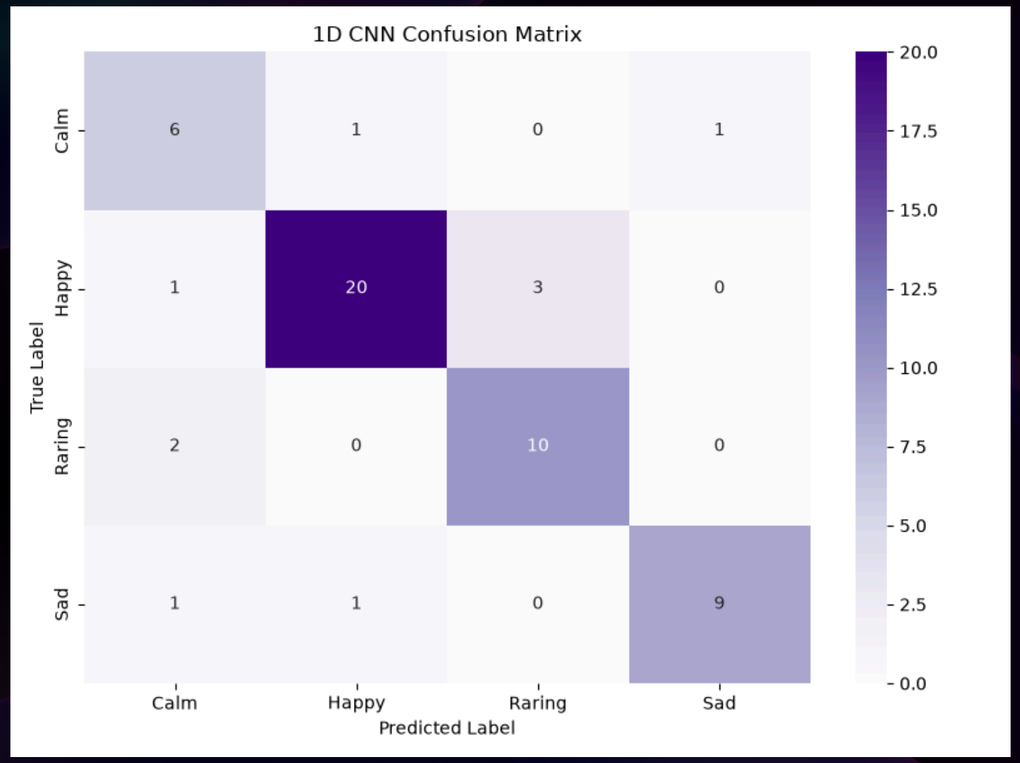


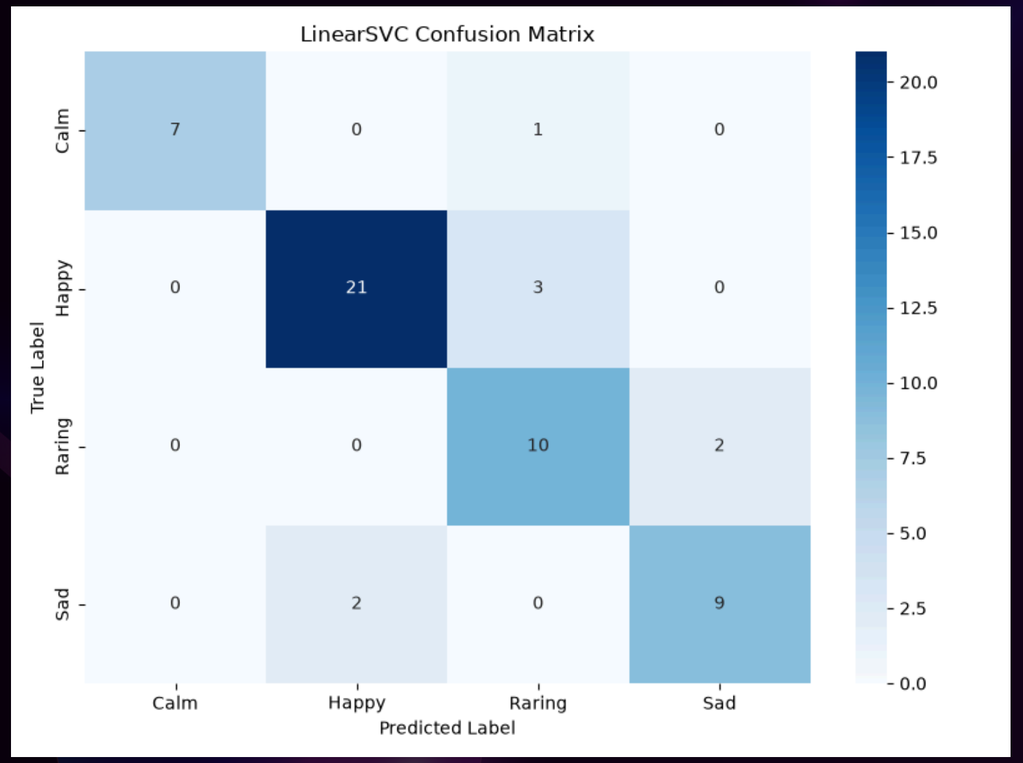
Shows the clean convergence of training and validation accuracy/loss, proving that early stopping successfully mitigated overfitting.

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ERROR ANALYSIS & CONFUSION MATRICES





This displays the confusion matrices to highlight strong diagonal classification accuracy.

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QUANTITATIVE COMPARISON W/ LITERATURE

STUDY/REFERENCE ARCHITECTURE	TASK DOMAIN	LANGUAGE SCOPE	METRIC/F1-SCORE	KEY LIMITATION
MONOLINGUAL BERT (DEVLIN ET AL.)	GENERAL SENTIMENT	ENGLISH ONLY	F1: ~0.78	Fails on code switched structures
MULTILINGUAL LSTM BASELINE (KIM ET AL.)	LYRIC CLASSIFICATON	MULTI-LANGUAGE	F1: ~0.42	High susceptibility to overfitting
REGULARIZED DUAL-INPUT CNN	EMOTION CLASSIFICATION	BILINGUAL (KOREAN-ENGLISH)	F1: 0.3354	Mitigated overfitting via L2 & Early Stopping

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COMPREHENSIVE ERROR ANALYSIS

Competitive DL Performance

The regularized Dual-Input 1D CNN successfully closed the performance gap, achieving 81.82% accuracy and closely matching the linear baseline while leveraging rich metadata fucion.

Methodological Rigor

Resolved previous structural bottlenexks through acoustic label grounding, rigorous data splitting, and complete literature benchmarking.

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CONCLUSION & DISCUSSION

- **Competitive Deep Learning Performance**
  - The regularized Dual-Input 1D CNN successfully closed the performance gap, achieving 81.82% accuracy and closely matching the linear basely while leveraging rich metadata fusion.
- **Methodologcal Rigor:**
  - Resolved previous structural bottlenexks through acoustic label grounding, rigorous data splitting, and complete literature benchmarking.